

AMIT PRAKHAR PANDEY

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Education

Indian Institute of Information Technology, Dharwad

Dec 2020 – May 2024

Bachelor of Technology in Computer Science and Engineering (CPI: 8.54)

Dharwad, Karnatak

Relevant Coursework

- Data Structures and Algorithms
- DBMS
- Computer Network
- Object Oriented Programming
- Operating System

Experiences

MediCodio

July 2024 – Present

Software Engineer Module Lead (Platform & Application)

- **Architected MediCodio's core AI engine** for medical code prediction, achieving **98%+ accuracy** across multiple specialties through optimized LLM workflows and advanced prompt engineering.
- Built scalable, modular pipelines capable of processing **large chart volumes** with future-proof architecture and robust edge-case handling.
- Reduced platform **execution time by 50%** via workflow optimization, vector-search enhancements, and efficient chart-processing pipelines.
- Cut operational costs by **75%** through optimized model calls, caching strategies, and RPA workflow improvements.
- Led the **Platform/Data Science team** in solving complex specialty-based prediction challenges, improving reliability and code coverage.
- Developed MediCodio's full application platform **from scratch using the PERN stack**, taking end-to-end ownership of frontend, backend, and database modules.
- Boosted application performance by **40%** through API optimization, React state-management restructuring, and PostgreSQL tuning.
- Delivered high-impact product features in **days**, accelerating development velocity and significantly reducing deployment cycles.
- Led the Application and Integration team, enabling seamless **API/RPA integrations with EMR systems** and resolving high-priority production issues.
- Communicated daily with **US-based clients** for requirement gathering, integration alignment, and successful production deployments.

Projects

Codio Prediction Engine – AI-Based Medical Code Prediction System | *LLMs, Prompt Engineering, Pinecone Vector DB,*

- * Designed and engineered MediCodio's **core AI prediction engine** responsible for end-to-end chart processing, enrichment, and code generation.
- * Achieved **95%+ prediction accuracy** across multiple specialties through specialty-aware workflows and optimized LLM reasoning.
- * Reduced **execution time by 50%** by improving vector search flows, batching, caching, and model-call strategies.
- * Cut operational cost by **75%** using advanced Prompt Engineering, fallback models, and optimized inference pipelines.
- * Built scalable, modular components capable of processing **large chart volumes** with strong edge-case handling and future scalability.

MediCodio Application Platform – AI & RPA Integrated B2B SaaS | *React.js, Node.js, Express, PostgreSQL, RPA, API In*

- * Built the **complete MediCodio application** from scratch using the PERN stack with full ownership of frontend, backend, and database.
- * Improved application performance by **40%** through API optimization, React state-management restructuring, and PostgreSQL tuning.
- * Developed secure REST APIs, coder dashboards, chart review interfaces, queue management, and workflow orchestration modules.
- * Integrated AI engine endpoints for **predictive coding**, real-time status updates, audits, and chart lifecycle management.
- * Delivered major product features in **days**, supporting a rapid development and deployment cycle.

End-to-End EMR Integration & Medical Chart Automation System | *RPA, API Integrations, Workflow Design*

- * Developed the **complete integration pipeline** for chart ingestion from multiple EMR systems using API and RPA-based automations.
- * Automated workflows for **chart import → validation → prediction → coder review → audit → export**, reducing manual operations by **90%**.
- * Implemented mandatory-field validation, denial detection, risk scoring, audit logic, and custom routing based on specialty.

- * Created fault-tolerant execution architecture with retries, fallback workflows, and detailed logging for compliance.
- * Collaborated with US-based clients to align data mapping, EMR setup, chart formats, and workflow execution.

Technical Skills

Languages/Frameworks: C, C++, Python, JavaScript, HTML/CSS, SQL, React.js, ExpressJS, Node.js

Developer Tools/Platforms: Visual Studio Code, Git/Github, Jupyter Notebook, Bootstrap, Tailwind CSS, Postman, Azure (Azure Services)

AI/ML Tools & Technologies: OpenAI API, LangChain, HuggingFace Transformers, NER, NumPy, Pandas, Scikit-learn, Cursor, N8N, Prompt Engineering & LLM Optimization, Generative AI (GenAI), Agentic AI Workflows

Databases: MongoDB, MySQL, PostgreSQL, Pinecone (Vector DB),

Additional Skills: Strong Problem-solving and Analytical skills, Effective Interpersonal Communication and Collaborative Teamwork skills

Achievements and Certifications

- Solved **600+** problems across different problem-solving platforms. **Link**
- Completed ReactJS course on Udemy. **Certificate Link**
- Completed HTML/CSS JavaScript course on Udemy. **Certificate Link**
- Completed Complete Introduction to C++ course on Udemy. **Certificate Link**